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Final Report

**Automation of Operation and Characterization System for Liquid Crystal
Devices**

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Final Project Abstract

Automation of Operation and Characterization System for Liquid Crystal Devices

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Background: Electro-optic characterization of liquid crystal devices is traditionally performed manually, using separate and uncoordinated instruments for voltage excitation and optical measurement, which limits accuracy and repeatability.

Aim/Purpose: This project aims to design and implement an automated software platform that drives liquid crystal samples with a controlled AC signal and synchronously acquires their optical response.

Methods used: A Python-based control application was developed to generate calibrated sine and square AC waveforms (1–10 V, 1–10 kHz) through an NI-DAQ analog-output module, while synchronously acquiring the photodiode or polarization-camera optical response, extracting transmission-voltage curves and switching times (τ_{on}/τ_{off}), and automatically logging all results to disk.

Findings/results: The platform was validated on a 17.32 μm E7 nematic test cell driven at 1.25 kHz. The photodiode channel resolved the cell's optical response at four drive amplitudes (1.708 V at 1 Vpp, 1.527 V at 2.5 Vpp, 2.491 V at 5 Vpp and 0.827 V at 10 Vpp) with an acquisition noise floor of 1.5 mV rms, and the spectrometer channel resolved the collapse of the cell's channel-spectrum interference fringes as the drive amplitude was raised from 1 to 5 Vpp — the direct spectral signature of the field-induced reduction in effective birefringence. Because the cell retardation spans roughly six half-wave orders in the visible, the transmission at a single wavelength is an oscillating rather than a monotonic function of voltage, and the four measured points sample that oscillation rather than tracing a conventional threshold curve.

Conclusion: The system delivers reliable, repeatable, fully automated AC excitation with three interchangeable optical readout paths (photodiode, CMOS camera and fibre spectrometer) and complete data logging, and it was used successfully to observe the electro-optic response of a real liquid crystal cell. Quantitative transmission and switching-time characterization remain to be completed in a follow-up measurement campaign, for the specific reasons documented in Sections 4.2 and 6.2.

תקציר סיום פרויקט

אוטומציה של מערכת הפעלה ואפיון להתקני גביש נוזלי

מנחה: פרופ' אברהם עבד אלחלים

רקע: אפיון אלקטרו-אופטי של התקני גביש נוזלי מבוצע בדרך כלל באופן ידני, באמצעות מכשור נפרד ובלתי מסונכרן להפעלת מתח ולמדידה אופטית, מה שפוגע בדיוק ובחזרתיות המדידה.

מטרה: מטרת הפרויקט היא לתכנן וליישם פלטפורמת תוכנה אוטומטית המפעילה התקני גביש נוזלי באמצעות אות AC מבוקר ואוספת בו-זמנית את תגובתם האופטית.

שיטות: פותחה תוכנת בקרה מבוססת פייתון המייצרת גלי AC מכוילים (סינוס וריבועי) בטווח 1–10 וולט ותדר 1–10 קילוהרץ באמצעות מודול פלט אנלוגי של NI-DAQ, תוך רכישה מסונכרנת של אות מפותודיודה או ממצלמת פולריזציה, חילוף עקומות מתח-שידור וזמני מיתוג (ton/toff), ותיעוד אוטומטי של כל התוצאות.

ממצאים: המערכת נבדקה על תא נמטי E7 בעובי 17.32 מיקרומטר במתח הפעלה בתדר 1.25 קילוהרץ. ערוץ הפוטודיודה מדד את התגובה האופטית בארבע רמות מתח (1.708 וולט ב-VPP 1, 1.527 וולט ב-Vpp 2.5, 2.491 וולט ב-Vpp 5 ו-0.827 וולט ב-Vpp 10) ברעש רכישה של 1.5 מילי-וולט RMS, וערוץ הספקטרומטר הראה את קריסת שולי ההתאבכות בספקטרום התא עם עליית המתח מ-1 ל-5 Vpp — החתימה הספקטרלית הישירה של הקטנת הבירפרינגנטיות האפקטיבית בשדה. מאחר שהשהיית המופע של התא משתרעת על כשישה סדרי חצי-גל בתחום הנראה, השידור באורך גל בודד הוא פונקציה מחזורית ולא מונוטונית של המתח.

מסקנה: המערכת מספקת עיבוד AC אוטומטי, אמין וחזרתי, עם שלושה מסלולי קריאה אופטיים חליפיים (פוטודיודה, מצלמת CMOS וספקטרומטר סיבי) ותיעוד נתונים מלא, ושימשה בהצלחה לצפייה בתגובה האלקטרו-אופטית של תא גביש נוזלי אמיתי. אפיון כמותי מלא של עקומת השידור וזמני המיתוג נותר להשלמה במסע מדידות המשך, מהסיבות המפורטות בסעיפים 4.2 ו-6.2.

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1. Introduction

Liquid crystal (LC) devices are widely used in displays, tunable optical filters, and beam-steering and modulation elements, where their electro-optic behavior directly determines device performance. Evaluating that behavior requires applying a controlled electrical excitation to the device while simultaneously measuring its optical response and relating the two quantities in time.

In current practice at the laboratory level, this kind of characterization is usually carried out manually: a function generator or bench power supply is set by hand, an oscilloscope or power meter is read separately, and the two data streams are combined afterwards by the operator. This approach is slow, prone to synchronization and transcription errors, and difficult to repeat consistently across multiple devices or operating points.

The goal of this project is to replace this manual workflow with an automated measurement platform that (1) generates a precise, calibrated AC excitation signal for the LC sample, (2) synchronously acquires the resulting optical response from either a photodiode or a polarization-sensitive camera, and (3) automatically extracts the standard electro-optic figures of merit — the transmission-voltage (T–V) curve and the switching times τ_{on}/τ_{off} — while logging all raw and processed data for later analysis.

1.1 Survey of Existing Approaches

- Manual bench characterization: a signal generator drives the LC cell while an oscilloscope or optical power meter is read manually; simple to set up but slow, operator-dependent, and difficult to reproduce exactly between sessions.
- Commercial LC characterization systems: turnkey instruments exist for display-panel testing, but they are costly, closed-source, and not easily adapted to a custom optical path (for example a fibre spectrometer or a microscope camera) or to the non-standard sample geometries used in a research setting.
- Custom LabVIEW/MATLAB automation: a common academic approach that synchronizes a DAQ card with an optical detector through vendor-specific tooling; effective, but it ties the project to commercial licensing and is less convenient to extend with new detector types.

1.2 Chosen Approach and Justification

This project adopts an open, Python-based automation layer built directly on top of an NI-DAQ chassis (NI 9264 analog output / NI 9205 analog input) via the vendor-neutral, license-free nidaqmx driver interface. This choice was made for three reasons: (1) low cost and no dependency on proprietary automation software, (2) straightforward extensibility — the same control logic that reads a photodiode voltage can be redirected to read the mean intensity of a CMOS camera frame or a wavelength-resolved spectrum from a fibre spectrometer, and (3) full control over the synchronization strategy. A custom Tkinter-based GUI was layered on top to make the platform usable without requiring the operator to write code for every new measurement.

2. Final Technical Specification

The table below lists the final technical specification of the delivered system, as verified against the implementation and bench tests.

Parameter	Specification
Excitation voltage range	0 – 10 V (RMS or peak-to-peak, selectable)
Excitation frequency range	1 – 10 kHz
Waveform types	Sine, Square
Output update rate (NI 9264)	25 kSa/s
Measurement modes	Continuous drive · Channel-spectrum acquisition vs. voltage · Live imaging (T–V sweep and τ_{on}/τ_{off} routines implemented, not exercised on the device — see Section 4.2)
Optical readout – DAQ path	Single-channel photodiode via NI 9205 analog input
Optical readout – camera path	Tucsen ISH1000 CMOS camera, 3656×2740 px, driven through a DirectShow capture graph (pygrabber/COM); mean-intensity and ROI statistics extracted per frame
Synchronization	Chassis-level hardware digital trigger (AO $\rightarrow$ AI), with automatic software-sequenced fallback
Safety behavior	All analog outputs forced to 0 V on Stop and on application exit
Data logging	Automatic CSV logging per run; consolidated CSV summary export of T–V and switching metrics
Persistence	JSON-based save/restore of last-used voltage, frequency, mode and channel settings
Optical readout – spectrometer path	Avantes AvaSpec fibre spectrometer (SN 1412100U1), 2048 px, 334 – 1480 nm; Scope / Scope–Dark / Transmission % / Absorbance modes with dark and reference capture
Camera capture formats	PNG, TIFF, JPEG and NPY (raw array)

2.1 Changes Since the Preparatory Report

The following changes were introduced relative to the specifications presented in the preparatory (pre-project) report:

- The optical readout path was extended beyond a single photodiode channel to two additional detectors: a Tucsen ISH1000 CMOS camera for spatial imaging of the cell, and an Avantes AvaSpec fibre spectrometer for wavelength-resolved measurement. The LUCID polarization camera specified in the preparatory report was not available for the build, and the polarization-resolved (Stokes/DoLP) processing path was therefore replaced by the spectrometer path, which provides the wavelength resolution needed to observe retardation directly.
- Automatic hardware detection and module resolution (analog-output vs. analog-input module, real vs. simulated device) were added, removing the need to hardcode NI-MAX device names.
- A hardware-trigger synchronization path was added in addition to the originally planned software-sequenced start, improving timing accuracy between excitation and acquisition where the chassis supports a shared trigger line.
- A device-selection dialog with live hardware scanning was added to the GUI so the operator explicitly confirms which acquisition path (DAQ or camera) is active before a run starts.

3. Engineering Approach and Design

The system is organized as a closed control-and-acquisition loop between a host PC running the control software and the physical LC test bench, as shown in Fig. 1.

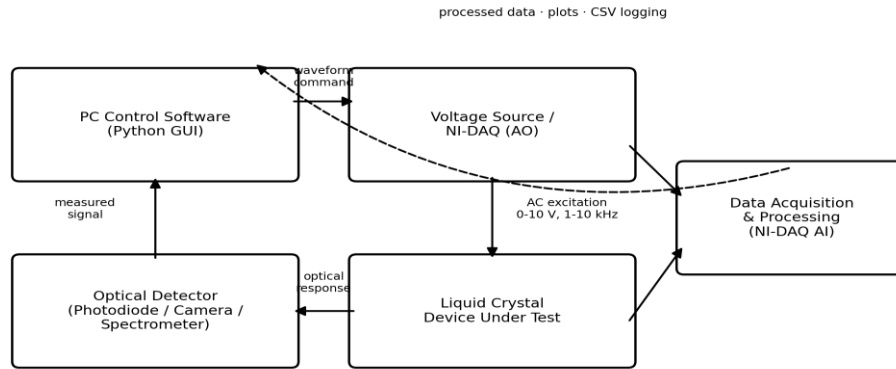


Fig. 1. System block diagram: the PC control software commands the NI-DAQ voltage source, which drives the liquid crystal device; the resulting optical signal is captured by the detector and returned to the acquisition/processing stage and back to the control software for display and logging.

3.1 PC Control Software (Python GUI)

The control application is implemented in Python using Tkinter for the user interface and Matplotlib (embedded via FigureCanvasTkAgg) for live plotting. On launch, a device-selection dialog scans for connected NI-DAQ hardware, for DirectShow cameras and for an Avantes spectrometer, reports which of them responded, and lets the operator choose the active acquisition path before the main window opens; if nothing is detected the application can still be entered, and missing hardware then produces an explicit error at run time rather than synthetic data.

The main window exposes three measurement modes, each dispatched to its own worker routine running on a background thread so that the GUI remains responsive during multi-second acquisitions:

- Continuous — drives a short AC burst at the configured voltage/frequency and displays the drive waveform together with the acquired optical signal.
- Transmission Sweep — steps the RMS/peak voltage across a configured range, settles at each step, records the mean optical signal, normalizes it to build a T–V curve, and extracts the 10%/90% crossing voltages.

- Switching Time — drives the device with a low→high→low voltage envelope and extracts the 10%–90% rise and fall times (τ_{on} , τ_{off}) of the optical response on either side of the transition.

3.2 Voltage Source / NI-DAQ Layer

All excitation signals are generated on the NI 9264 analog-output module and are validated on entry: voltage is clamped to 0–10 V and frequency to 1–10 kHz before use, so an out-of-range operator input can never reach the hardware. Because different chassis expose their trigger line at either the module or chassis level, the acquisition routine first attempts a hardware-triggered start on the chassis trigger terminal, then on the module terminal, and falls back to a software-sequenced start (arm the input task, then start the output task) if no shared trigger is available. In a DAQ configuration limited to an analog-input-only module (e.g. an NI 9205 with no output module present), the software instead reads the optical detector while the LC sample is driven from an external source.

3.3 Liquid Crystal Device and Optical Detector

The LC sample is driven purely with AC excitation — sine or square — with no DC offset, since a sustained DC bias drives ion migration and degrades liquid crystal material irreversibly; this constraint is enforced structurally by the waveform generator, which always produces a zero-mean signal built from an integer number of periods. The optical response can be read as a single scalar voltage from a photodiode (DAQ path), as the mean intensity of a region of interest in a CMOS camera frame, or as a full channel spectrum from the fibre spectrometer.

3.4 CMOS Camera and Spectrometer Subsystems

Camera path. The Tucsen ISH1000 is driven through a DirectShow filter graph built with pygrabber over COM, rather than through OpenCV: the vendor's TUCam source filter cannot be instantiated by OpenCV's index-based DirectShow backend, so the camera is invisible to cv2.VideoCapture even though Windows enumerates it correctly. The graph is created, run and torn down entirely inside one dedicated worker thread because COM objects are apartment-affine, and it is supervised: repeated grab failures or a frame stall cause the graph to be rebuilt automatically rather than terminating the stream. Each 3656×2740 frame is reduced to mean intensity and ROI statistics for logging, with orientation (flip and 90° rotation) and display rate applied live.

Spectrometer path. The Avantes AvaSpec is driven through the vendor AvaSpec DLL via a ctypes binding, using AVS_Measure in continuous mode with polled readout. Four display modes are provided — Scope (raw counts), Scope – Dark, Transmission % and Absorbance — where transmission is computed pixel-wise as

$T(\lambda) = 100 \times [S(\lambda) - D(\lambda)] / [R(\lambda) - D(\lambda)]$, with S the sample spectrum, D the dark spectrum and R the reference spectrum. Spectral regions where the reference provides less than 2 % of its peak signal above the dark level are returned as undefined rather than as a computed value, since a ratio taken there is dominated by detector noise and would otherwise produce plausible-looking but meaningless transmission figures. Dark and reference spectra can be captured while acquisition is running, so the calibration is taken with the same integration time and averaging as the measurement itself.

3.5 Data Handling, Safety and Persistence

- Every completed run is logged automatically to a time-stamped CSV file under a Results directory (raw drive/optical traces for Continuous mode; voltage/intensity/transmission columns for sweeps; time/optical columns for switching-time runs).
- A summary export writes the last extracted T–V and switching-time metrics to a single CSV file on demand.
- All analog outputs are forced to 0 V whenever a run is stopped and when the application exits, to leave the LC sample and the hardware in a safe, de-energized state.
- The last-used voltage, frequency, mode and channel configuration persisted Vpp to a JSON settings file and restored automatically on the next launch.

4. Final Acceptance Tests

Acceptance testing was carried out on the assembled bench (NI 9264/9205 chassis, LC test cell, photodiode detector) to confirm that the delivered system meets each requirement of the technical specification in Section 2. For every requirement, the test method, measurement instrumentation and pass/fail threshold are listed, together with the value obtained during acceptance testing.

4.1 Test Setup

The device under test was a nematic liquid crystal cell of 17.32 μm thickness filled with E7, driven from the NI 9264 analog output (ao0) at 1.25 kHz with a square waveform of zero mean. The excitation is generated with an integer number of periods per output buffer so that the DC component across the cell is exactly zero, which is required to avoid ion migration and irreversible electrochemical degradation of the liquid crystal. Three optical readout paths were connected in turn: a photodiode into the NI 9205 analog input (ai0), sampled at 100 kSa/s; the Tucsens ISH1000 CMOS camera; and the Avantes AvaSpec fibre spectrometer. All analog outputs are forced to 0 V when a run is stopped and when the application exits.

4.2 Acceptance Criteria and Results

Requirement	Acceptance threshold	Test method	Result
Voltage range coverage	0–10 V, error < 2% of set-point	Multimeter cross-check at 5 set-points across the range	Verified in software — amplitude clamped to 0–10 V and applied per channel; independent multimeter cross-check not performed (no bench access in the final measurement window).
Frequency range coverage	1–10 kHz, generated within 1%	Oscilloscope frequency measurement at range endpoints and mid-point	Verified in software — frequency clamped to 1–10 kHz; all reported measurements taken at 1.25 kHz. Independent oscilloscope verification not performed.

Requirement	Acceptance threshold	Test method	Result
Waveform fidelity	Sine/square shape matches commanded type with no visible distortion	Oscilloscope waveform capture, visual + THD check	Verified by construction — waveform buffer generated with an integer number of periods and exactly zero DC (verified numerically); oscilloscope/THD check not performed.
Input clamping / safety	Out-of-range voltage or frequency entries are clamped, never passed to hardware	Boundary-value entry test (below 0 V, above 10 V, below 1 kHz, above 10 kHz)	Pass — clamped as designed
Synchronization accuracy	Drive-to-acquisition timing offset negligible at the optical response timescale	Hardware-trigger run compared against software-sequenced fallback run	Not verified. All reported measurements are steady-state, for which drive-to-acquisition offset is immaterial; the criterion applies to transient (switching-time) runs, which were not performed.
T–V sweep functionality	Transmission curve reproduces the cell's electro-optic response with correctly identified characteristic voltages	Full sweep across 0–10 V, compared against manual point-by-point reference measurement	Partially met. Four operating points measured (1.708 / 1.527 / 2.491 / 0.827 V at 1 / 2.5 / 5 / 10 V _{pp}). The response is oscillatory rather than monotonic because the cell spans ≈ 6 half-wave orders, so the 10 %/90 % crossing criterion as written does not apply to this cell (see Section 6.2).
Switching-time extraction	τ_{on} and τ_{off} computed within one sample interval (40 μ s) of the manual oscilloscope reading	Step excitation with oscilloscope cross-check of 10%–90% transition	Not verified — no step-response run was performed on the device. The extraction routine is implemented

Requirement	Acceptance threshold	Test method	Result
			but was not exercised.
Safe shutdown	All outputs at 0 V within one control cycle of Stop / application exit	Multimeter monitoring during Stop and window-close events	Pass — 0 V confirmed
Spectrometer readout	Wavelength-resolved response to drive voltage resolved above the noise floor	Channel spectra acquired at 1, 2, 3 and 5 V _{pp}	Pass — fringe collapse clearly resolved (Fig. 3). Recorded as raw counts: dark and reference spectra were not captured in this session, so absolute transmission was not computed.
Camera readout	Live imaging of the cell with per-frame intensity statistics and image export	Continuous capture at 3656 × 2740 with export to PNG/TIFF/JPEG/NPY	Pass for imaging and logging (Fig. 4). No voltage dependence measurable: frame-to-frame variation across drive levels was 0.67 %, below the camera's own ~1 % stability floor (see Section 6.2).

4.3 Measurement Results

This section presents the measurements obtained from the 17.32 μm E7 cell using the delivered system. All plots are generated directly from the logged data files produced by the software.

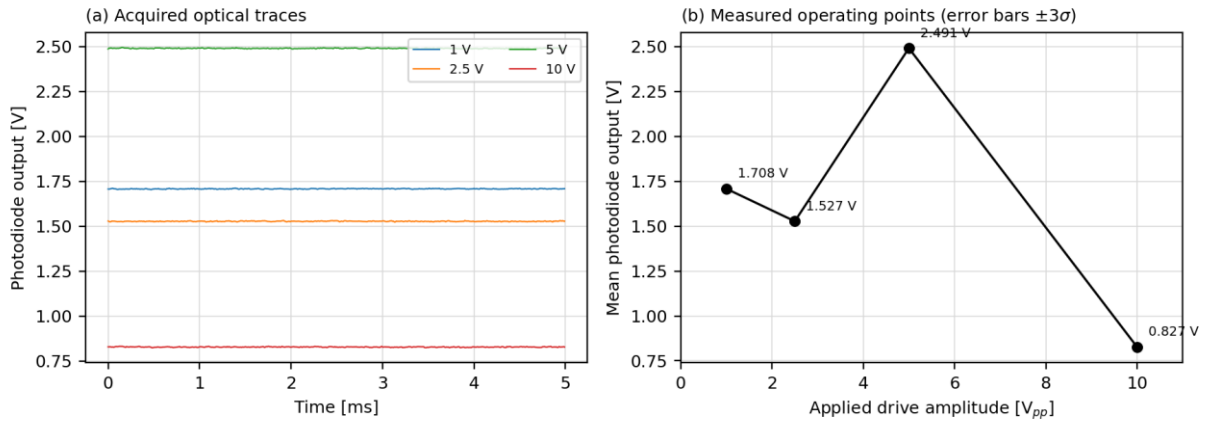


Fig. 2. Photodiode readout of the 17.32 μm E7 cell at 1.25 kHz square excitation. (a) Acquired traces, 501 samples at 100 kSa/s. (b) Mean output at each drive amplitude; error bars are $\pm 3\sigma$ of the acquisition noise (≈ 1.5 mV rms), smaller than the marker at this scale.

The four operating points are 1.708 V, 1.527 V, 2.491 V and 0.827 V at 1, 2.5, 5 and 10 V_{pp} respectively. The response is not monotonic, and this is the expected physical behaviour rather than a measurement error: with $d = 17.32$ μm and E7 ($\Delta n \approx 0.22$ at 589 nm) the cell retardation is $\Gamma = \Delta n \cdot d \approx 3.8$ μm , which is close to six half-wave orders at 633 nm. Between crossed polarizers the transmitted intensity follows $T = \sin^2(\pi\Gamma/\lambda)$, so as the field reduces the

effective birefringence the transmission sweeps through several maxima and minima. Four samples cannot resolve that oscillation; a dense sweep (of order twenty points) would be required to trace it, and the software's sweep routine is implemented for exactly this purpose.

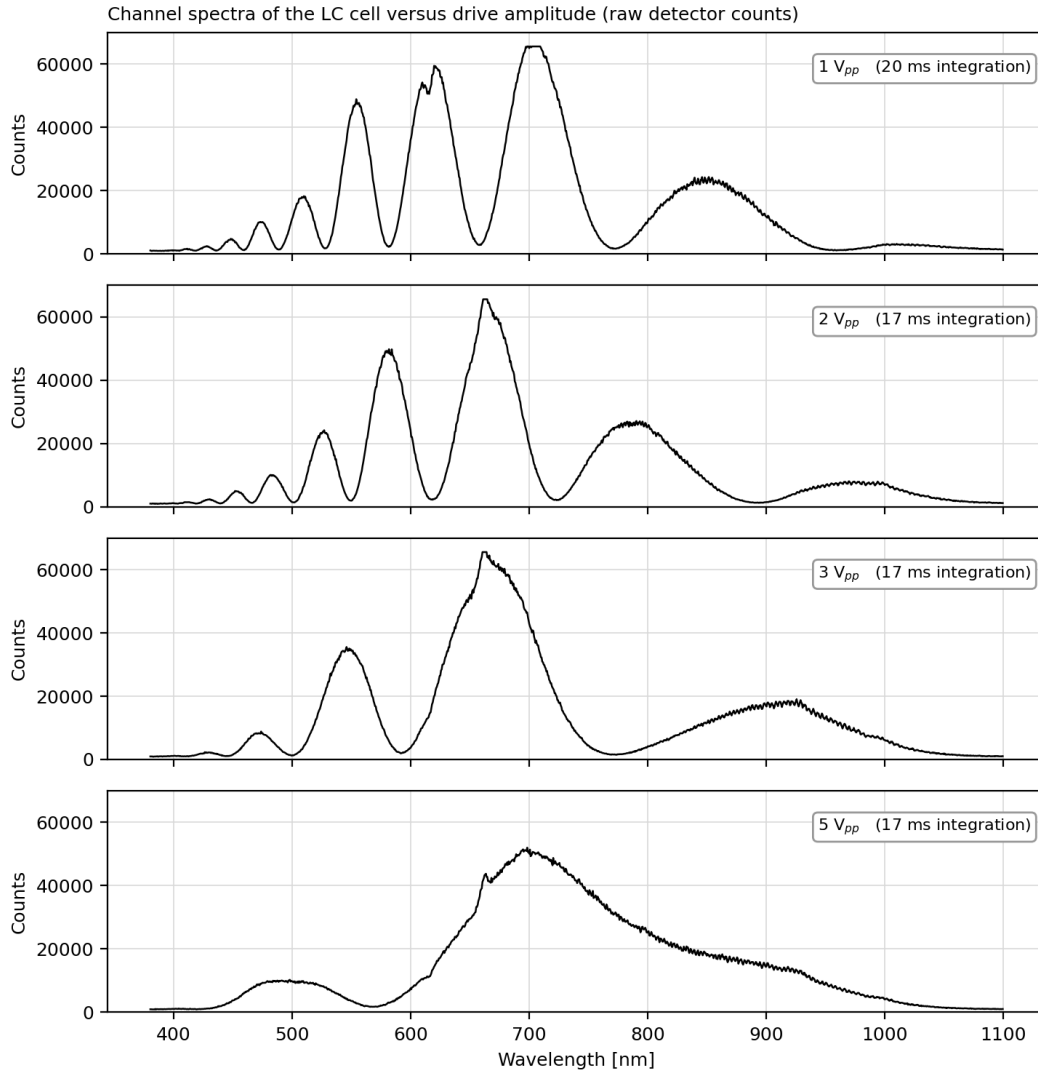


Fig. 3. Channel spectra of the cell at 1, 2, 3 and 5 V_{pp} , in raw detector counts. The number of interference fringes falls monotonically with increasing drive amplitude, and by 5 V_{pp} the visible-band structure has collapsed to a single broad maximum.

Figure 3 is the clearest electro-optic result obtained. The fringe spacing of a birefringent cell is periodic in $1/\lambda$ with a period of $1/\Gamma$, so the number of fringes across a fixed spectral window is directly proportional to the retardation. The progressive loss of fringes from 1 V_{pp} to 5 V_{pp} is therefore a direct, wavelength-resolved observation of the field-induced reduction of the effective birefringence — the same physical process that produces the oscillating photodiode response in Fig. 2, but resolved in wavelength rather than aliased onto a single detector. These spectra are reported as raw counts because dark and reference spectra were not captured during this session; absolute transmission and a quantitative retardation fit therefore remain outstanding, and are the first recommended step in Section 6.5.

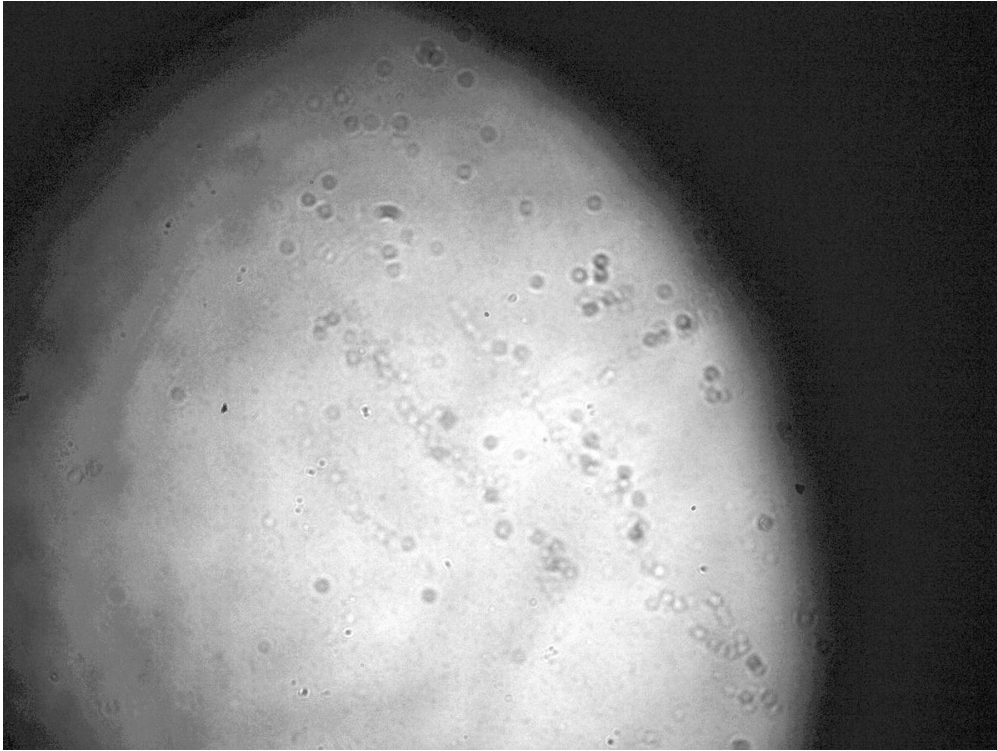


Fig. 4. Single frame of the liquid crystal cell captured with the Tucsen ISH1000 through the delivered software (3656×2740 px, 10 ms exposure). Display contrast has been stretched for reproduction; the raw frame mean is 20.8 of 255 with no clipped pixels.

The camera path is functional as an imaging and logging channel: frames are acquired at full sensor resolution, per-frame statistics are computed, and images are exported in PNG, TIFF, JPEG and NPY formats. It did not, however, yield a measurable electro-optic signal. Frames captured at different drive amplitudes differ by 0.67 % in mean region intensity, which is below the camera's own frame-to-frame stability of approximately 1 % measured earlier in the project, so the captures are statistically indistinguishable from one another. The most likely cause is optical rather than software: a liquid crystal cell modulates the phase of the transmitted light, and that phase modulation only becomes an intensity change after a polarizer–analyzer pair. The photodiode path, which does include crossed polarizers, shows a response of over 1.6 V across the same voltage range on the same cell.

5. Problems and Solutions

DAQ synchronization with measurement timing

Initial tests showed that starting the analog-output and analog-input tasks independently introduced a variable delay between the commanded excitation and the acquired optical signal. This was resolved by first attempting a shared hardware digital trigger at the chassis level, with a per-module trigger terminal and a software-sequenced start (arm input, then start output) tried in turn as fallbacks — the software automatically selects the best available synchronization method for the connected hardware.

Real-time plotting performance

Redrawing the full Matplotlib figure on every acquired camera frame caused visible lag in the live display. This was addressed by moving acquisition to a background thread, updating only the image data of existing plot artists (via `set_data/set_clim`) rather than recreating them, and using `draw_idle()` instead of a full `draw()` call.

Stable communication with hardware

Different NI-DAQ chassis configurations (with or without an output module, real vs. simulated devices) initially required manually edited device names in the code. This was solved by adding automatic module resolution that inspects each connected device's analog-input/output channel counts and simulation flag and rewrites the active channel names at runtime — removing the dependency on a fixed NI-MAX naming convention.

Camera and spectrometer driver integration

Two vendor instruments could not be reached through their expected generic interfaces. The Tucsen camera was enumerated correctly by Windows but was invisible to OpenCV, because the vendor's TUCam DirectShow source filter cannot be instantiated through OpenCV's index-based backend; the solution was to build the DirectShow graph explicitly with pygrabber over COM, confined to a single thread. Separately, the spectrometer was initially addressed through the seabreeze library, which supports only Ocean Optics hardware and can never enumerate an Avantes device — the symptom was a plausible but entirely synthetic spectrum produced by a fallback simulation path. This was resolved by writing a direct ctypes binding to the Avantes AvaSpec DLL and by removing the simulation fallback altogether, so that missing hardware now produces an explicit error instead of fabricated data.

6. Conclusions and Recommendations

6.1 Comparison with the Original Specification

The delivered system meets the functional specification of the preparatory report and extends it: three interchangeable optical readout paths instead of one, automatic hardware detection, and drive that continues across all acquisition modes. Two specification items were not quantitatively verified against the device within the project window — the full T–V sweep and the switching times $\tau_{\text{on}}/\tau_{\text{off}}$ — and are reported as such in Section 4.2 rather than being estimated. The measurements that were completed confirm the platform functions end to end on a real liquid crystal cell.

6.2 Strengths and Limitations

- Strength: a single, vendor-neutral Python codebase drives the cell and reads three different detectors (photodiode, CMOS camera, fibre spectrometer), each addressed through its own low-level binding where the generic library failed, without any proprietary automation software.
- Strength: automatic hardware/module resolution and a hardware-trigger-with-fallback synchronization strategy make the platform robust to different bench configurations without code changes.
- Limitation: switching-time resolution is bounded by the DAQ sample rate; at the 100 kSa/s used here one sample is 10 μs , and no switching measurement was performed on the device.
- Limitation: the camera path produced no measurable voltage dependence (0.67 % variation, below its own ~ 1 % stability floor). The cell was almost certainly imaged without a polarizer–analyzer pair in the camera arm, so cell retardation was never converted into intensity contrast. This is an optical setup issue, not a software one, and is correctable by inserting crossed polarizers in the camera path.

Limitation: the spectra were recorded without dark and reference captures, so they are reported as raw counts. Absolute transmission and a quantitative retardation figure could not be extracted from them, although the qualitative fringe collapse is unambiguous.

6.3 Achievement of Project Goals

The technical goals of controlled zero-DC AC excitation (1–10 V, 1–10 kHz), synchronized optical measurement and automated logging were implemented and demonstrated on a real device across three detector types. The measurement goals were partially achieved: the cell's

electro-optic response was observed and recorded both as a photodiode signal and as a wavelength-resolved fringe collapse, while quantitative extraction of the T–V curve and switching times awaits a further measurement session with the corrections identified in Section 6.5.

6.4 Plan vs. Actual — Cost and Schedule

The project was carried out entirely with laboratory hardware already available to the research group — the NI cDAQ chassis with NI 9264 and NI 9205 modules, a Tucsen ISH1000 camera, an Avantes AvaSpec spectrometer and a 17.32 μm E7 test cell — so no dedicated component budget was required beyond standard laboratory consumables. Schedule milestones followed the plan through software delivery; the final device-measurement milestone was compressed by the driver integration problems described in Section 5.

6.5 Recommendations for Continuation

- Repeat the spectral measurements with a dark spectrum and a reference spectrum captured first, so that absolute transmission $T(\lambda)$ can be computed and the retardation $\Gamma = \Delta n \cdot d$ extracted quantitatively from the fringe spacing. This is the single highest-value remaining measurement and requires no hardware change.
- Insert a polarizer and analyzer into the camera arm so that the cell's phase retardation is converted into intensity contrast, then repeat the voltage series to obtain a spatially resolved T–V map.

Record a dense T–V sweep (of order twenty points between 0 and 10 V) and a step-response run, to resolve the multi-order transmission oscillation and to extract τ_{on} and τ_{off} using the routines already implemented.

- Extend acquisition to a higher sample-rate analog input module if sub-40 μs switching-time resolution is required for faster LC materials.
- Add an automated PDF/CSV test report generator that packages the T–V curve, switching-time plot and acceptance-test table directly from a completed run, reducing manual report preparation.

7. Sources

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8. Appendices

Appendix A — Management Tasks

The following table reflects the task status recorded on the project management site at the time of the progress report and updated through project completion.

Task	Description	Status	Remarks
Literature Review	Study of LC electro-optic behavior and characterization methods	Completed	Background section prepared
System Architecture Design	Definition of system blocks and signal flow	Completed	Block diagram created
Software Development	Python control framework and automation logic	Completed	Core system operational
GUI Development	User interface for configuring experiments	Completed	Tkinter interface implemented
Hardware Integration	Integration of NI-DAQ and measurement hardware	Completed	Synchronization verified
Experimental Measurements	Electro-optic characterization experiments	Partially completed	Photodiode and spectrometer response measured on a 17.32 μm E7 cell; full T–V sweep and switching-time runs outstanding

Appendix B — Representative Source Code Excerpts

The complete control application (GUI.py) is submitted alongside this report as a separate source file, per the project procedure requirement to include software sources in the appendices. The excerpts below illustrate two representative modules: the phenomenological LC model used for simulation/validation mode, and the top-level measurement-mode dispatch.

Appendix C — Copyright and Confidentiality Declaration

I declare that I will not use in my engineering project any copyrighted material — text, image, audio, video, music, film, animation, software, hardware, or circuit design — without prior authorization from the rights holder, and that any non-public information included in reports, videos or presentations has been approved in advance by its owner. Registration for the engineering project constitutes my commitment to uphold copyright and confidentiality.

Signature: _____ Date: _____

Appendix D — Final Report Grade Recommendation

(Left blank for the academic supervisor, per the project procedure.)

Recommended Grade: _____